



Class 11th Physics — Chapter 6

Work, Energy & Power

Complete NCERT Notes for Class 11 Physics

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CHAPTER OVERVIEW

What is Chapter 6 About?

Chapter 6 introduces the concepts of **Work, Energy, and Power** — three fundamental quantities that describe how forces cause motion and change in physical systems.

⚡ Work

Force causing displacement

🔋 Energy

Capacity to do work

⚙️ Power

Rate of doing work

🔄 Conservation

Energy is never lost

Work Done by a Force



Definition

Work is said to be done when a **force** acts on an object and the object undergoes **displacement** in the direction of the force.

Formula

$$W = F \cdot d \cdot \cos \theta$$

Where **F** = Force, **d** = Displacement, **θ** = Angle between force and displacement.

① **SI Unit:** Joule (J) | **Scalar Quantity**

Types of Work Done

✓ Positive Work

When $\theta = 0^\circ$, force and displacement are in the **same direction**. Example: A ball falling freely under gravity.

$$W = Fd \cos 0^\circ = +Fd$$

✗ Negative Work

When $\theta = 180^\circ$, force and displacement are in **opposite directions**. Example: Friction opposing motion.

$$W = Fd \cos 180^\circ = -Fd$$

○ Zero Work

When $\theta = 90^\circ$, force is **perpendicular** to displacement. Example: Centripetal force in circular motion.

$$W = Fd \cos 90^\circ = 0$$

Energy — Kinetic & Potential

Kinetic Energy (KE)

Energy of an object **due to its motion**.

$$KE = \frac{1}{2}mv^2$$

Where **m** = mass, **v** = velocity

Potential Energy (PE)

Energy of an object **due to its position** or configuration.

$$PE = mgh$$

Where **m** = mass, **g** = gravity, **h** = height



Work-Energy Theorem

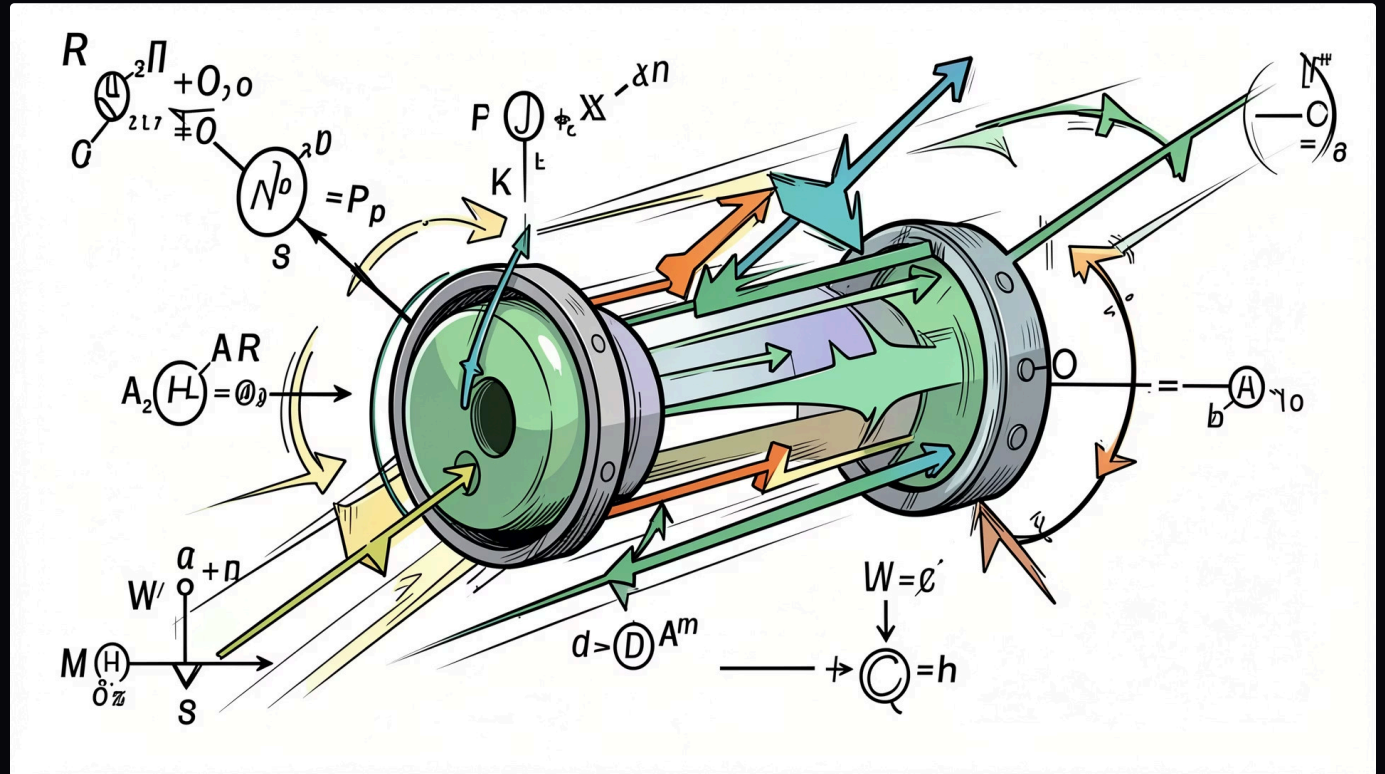
Statement

The **net work done** on an object is equal to the **change in its kinetic energy**.

$$W_{net} = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Key Points

- Applies to both constant and variable forces
- Derived from Newton's Second Law
- Work done **increases** KE; negative work **decreases** KE



Conservation of Energy

Energy can neither be created nor destroyed — it can only be **transformed** from one form to another.



Free Fall Example

PE converts to KE. Total energy (PE + KE) remains **constant** at all points.



Pendulum Motion

At extremes: max PE, zero KE. At mean position: max KE, zero PE.

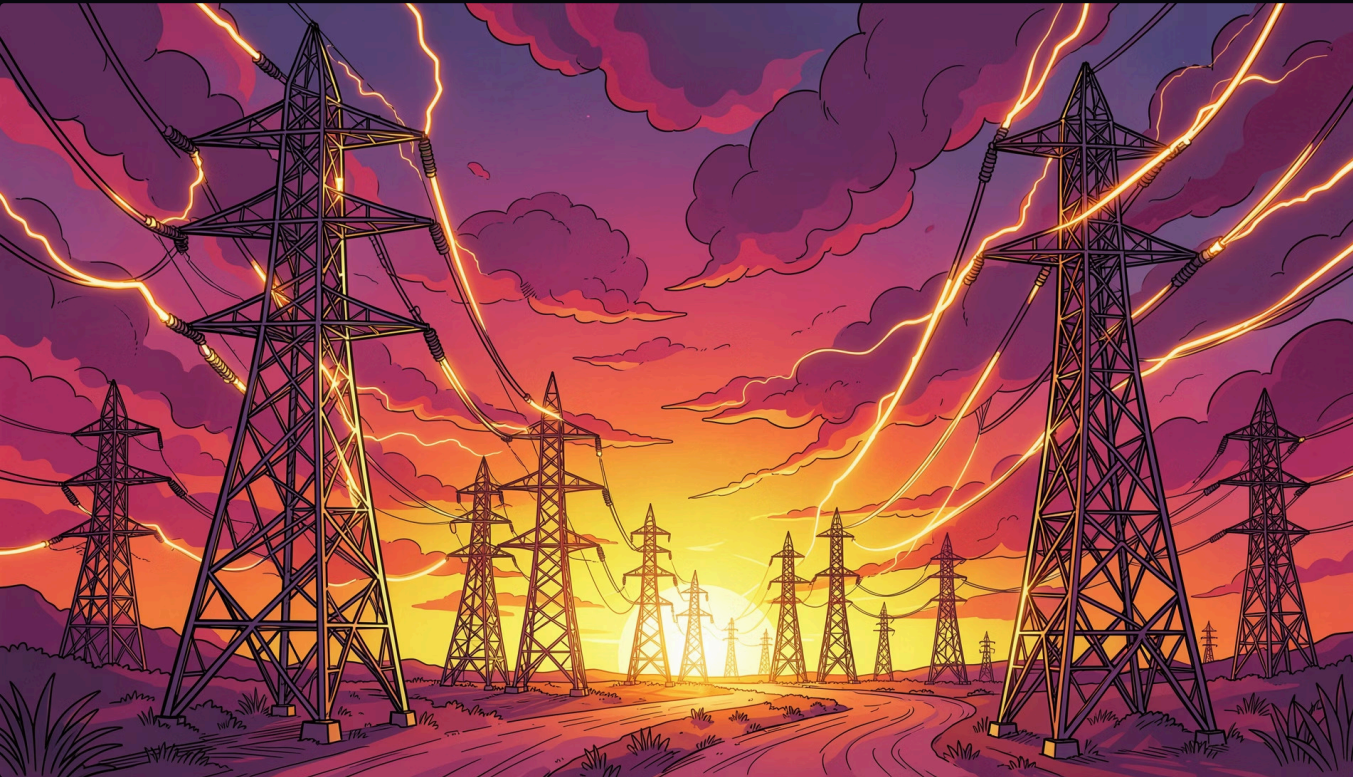


Spring System

Elastic PE ↔ KE conversion. Total mechanical energy is **conserved**.



Power



Definition

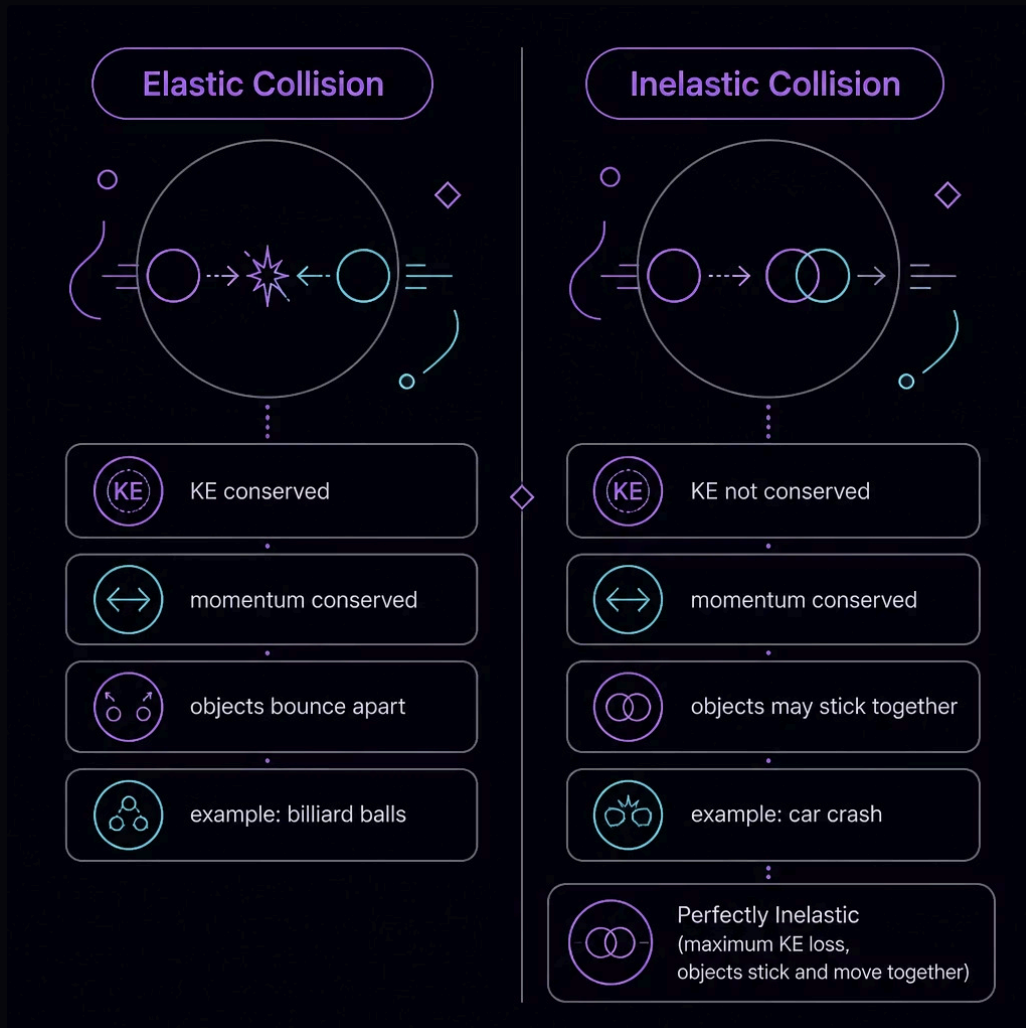
Power is the **rate of doing work** or rate of energy transfer.

$$P = \frac{W}{t} = F \cdot v$$

Units & Types

- **SI Unit:** Watt (W) = Joule/second
- **Commercial Unit:** Kilowatt-hour (kWh)
- **1 HP = 746 W**
- Instantaneous Power: $P = F \times v$

Collisions — Elastic & Inelastic



Key Formulas

Conservation of Momentum:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Coefficient of Restitution (e):

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

- Elastic: **e = 1**
- Inelastic: **0 < e < 1**
- Perfectly Inelastic: **e = 0**

SUMMARY

Chapter 6 — Quick Revision

01

Work

$W = Fd \cos \theta$ — Positive, Negative, Zero

02

Energy

$KE = \frac{1}{2}mv^2$ | $PE = mgh$ | Work-Energy Theorem

03

Conservation

Total Mechanical Energy is conserved in isolated systems

04

Power

$P = W/t = Fv$ — SI Unit: Watt

05

Collisions

Elastic ($e=1$) vs Inelastic ($e<1$) — Momentum always conserved



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